Physics-Informed Accelerated MRI Reconstruction with SwinUNet: Auditing, Correcting, and Extending a Transformer Reconstruction Pipeline

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Abstract

Magnetic Resonance Imaging provides detailed anatomical information without ionising radiation, but its long acquisition times limit throughput and patient comfort. Accelerated MRI addresses this by undersampling k-space and reconstructing the missing information computationally. In earlier work I proposed a hybrid U-Net/Swin Transformer architecture for this task and reported that the SwinUNet variant reached 33.10 dB PSNR on the fastMRI single-coil knee benchmark, above a U-Net baseline at 28.03 dB.

This paper reports a systematic audit of the released implementation and a substantially revised system. The audit found that the released code could not execute: the SwinUNet could not be *constructed* at its own documented configuration, because window partitioning assumed feature-map resolutions divisible by the window size and the third encoder stage is 20×20 with a window size of 8. Two further defects made every inference and evaluation entry point raise on invocation. A second class of defects ran silently but invalidated measurements: exponential-moving-average weights were never loaded, the data-consistency module was never given data and was therefore a complete no-op, the equispaced sampling mask acquired 19.3 % more k-space lines than its nominal acceleration, and PSNR was averaged over mini-batches before the logarithm, which makes the metric depend on batch size.

I rebuild the system around the physics of the acquisition. Reconstruction is performed on complex-valued data through an unrolled cascade in which every learned refinement is followed by a data-consistency projection that restores the measured Fourier coefficients exactly, verified here to 8.3×10^{-7}. Metrics are computed per image against each target's own dynamic range, and model comparisons report bootstrap confidence intervals and Holm-corrected paired permutation tests rather than point estimates.

In a controlled ablation on synthetic phantom data, the data-consistency cascade beats a capacity-matched single-pass network by 2.40 dB on a U-Net backbone and 2.16 dB on a SwinUNet backbone, both at $p < 10^{-4}$ under a paired permutation test and both with fewer parameters than the control. The near-equality of the two gains across architectures with very different inductive biases indicates the improvement comes from the physics constraint rather than the network. A secondary finding is that complex-valued input on its own *costs* 2.14 dB, because the magnitude loss leaves phase unsupervised — which is precisely the freedom the data-consistency projection removes. I do not restate the previously reported fastMRI figures as validated results: they are not reproducible from the released artifact, and the corrections to the sampling and metric definitions mean that numbers produced under the old protocol are not comparable to numbers produced under the new one. The revised system, its 267-test suite, and a synthetic data generator that makes the full pipeline

runnable without the fastMRI download are released so that the benchmark numbers can be regenerated under a protocol that is stated precisely and checked automatically.

Contents

1 Introduction

Magnetic Resonance Imaging (MRI) is among the most valuable non-invasive diagnostic modalities, offering exceptional soft-tissue contrast without ionising radiation. Its principal limitation is time. A fully sampled acquisition requires the scanner to traverse k-space line by line, which keeps patients immobile for extended periods, reduces scanner throughput, invites motion artefacts, and raises cost.

Accelerated MRI acquires only a fraction of k-space and recovers the remainder computationally. Undersampling below the Nyquist rate makes the inverse problem ill-posed: the naive zero-filled reconstruction exhibits coherent aliasing that obscures precisely the fine structures—cartilage boundaries, meniscal tears, cortical margins—that determine diagnostic value. Classical remedies include parallel imaging (SENSE [11], GRAPPA [3]) and compressed sensing [10]. Deep learning has since become the dominant approach, and the fastMRI benchmark [14] has made progress measurable.

1.1 Motivation for this paper

My earlier study compared a baseline U-Net, a U-Net with a Transformer at the bottleneck (BT-UNet), and a SwinUNet, and reported that the SwinUNet reached 33.10 dB PSNR and 0.7274 SSIM, against 28.03 dB and 0.6935 for the U-Net. That gap is 5.07 dB, though the prose of that paper described it as "approximately 4 dB" in three places—the first and mildest of the inconsistencies this audit uncovered.

Preparing the work for release, I audited the implementation against its own documentation. The audit did not find the small discrepancies one expects. It found that the artifact could not run at all, and that several components which appeared to be working were not.

The most consequential finding concerns the flagship model. With the documented configuration—320 px input, patch size 4, three encoder stages, window size 8—the stage resolutions are $80 \rightarrow 40 \rightarrow 20$. Window partitioning was implemented as a reshape,

```
x = x.view(B, H // ws, ws, W // ws, ws, C)
```

which silently requires H and W to be exact multiples of `ws`. At the third stage $20 \bmod 8 \neq 0$, and because the shifted-window attention mask was precomputed in `__init__`, the failure occurred at *construction*:

```
RuntimeError: shape '[1, 2, 8, 2, 8, 1]' is invalid for input of size 400
```

No forward pass through the reported architecture was possible from the released code. A separate misspelled module filename (`reproductibility.py`, imported as `reproducibility`) made `import training` raise `ModuleNotFoundError`, and two methods on the inference pipeline were missing their `@classmethod` and `@staticmethod` decorators, so every documented inference, benchmarking and export command raised immediately.

That a codebase in this state passed its own continuous-integration suite has a mundane explanation that is worth recording: the workflow file was at `.github/.workflows/ci.yml`. GitHub reads only `.github/workflows/`, so CI had never run. The repository's architecture-sanity command would not have caught it either—it wrapped each forward pass in a bare `except`, printed `FAIL` into a formatted table, and exited with status 0.

1.2 Contributions

1. **A reproducibility audit** of the released implementation, classifying 19 defects into those that prevent execution, those that run but silently disable a documented feature, and those that corrupt measurement. Each is characterised by mechanism and, where it affects reported numbers, quantified (Section 3).
2. **A corrected and generalised SwinUNet**. Feature maps are padded to the window size with padded tokens masked out of attention, so any resolution and any window size are valid; attention masks are built lazily and cached per resolution rather than baked in at construction. The architecture gains stochastic depth, truncated-normal initialisation, fused attention, and a zero-initialised residual head that makes the network the exact identity at step 0 (Section 5.2).
3. **A physics-informed reconstruction pipeline**. Reconstruction operates on complex data through an unrolled cascade of learned refinements interleaved with data-consistency projections. I show the projection restores measured coefficients to 8.3×10^{-7} (Sections 5.3 and 7.1).
4. **A corrected simulation and evaluation protocol**: sampling masks that achieve their nominal acceleration, augmentation applied before undersampling so every training sample remains a realisable acquisition, per-image metrics against each target's own dynamic range, and comparisons reported with bootstrap confidence intervals and Holm-corrected paired permutation tests (Sections 4 and 6).
5. **A controlled ablation** on synthetic phantom data with capacity-matched controls, isolating the contribution of complex-valued input and of the data-consistency cascade from that of parameter count (Section 7.2).
6. **A verifiable artifact**: 267 automated tests at 88 % statement coverage, including exactness assertions on the Fourier and data-consistency operators, schema validation that rejects invalid configurations before training starts, and a synthetic data generator that makes the entire pipeline runnable without the 90 GB fastMRI download (Section 8).

1.3 What this paper does not claim

I do not report validated fastMRI benchmark results. The previously published figures cannot be reproduced from the released code, and the corrections described here—to the sampling mask, the metric definition, and the augmentation ordering—change what those numbers measure, so old and new results are not comparable. Producing new fastMRI numbers requires GPU-scale training runs that are outside the scope of this paper. What is provided instead is a system in which such numbers can be generated under a protocol that is stated precisely, validated automatically, and reproducible from a recorded commit. Section 8.3 gives the exact commands.

2 Related Work

2.1 Traditional reconstruction

Parallel imaging exploits redundancy across receiver coils: SENSE unfolds aliasing in image space using coil sensitivity maps [11], while GRAPPA interpolates missing k-space lines from autocalibration data [3]. Compressed sensing instead exploits sparsity in a transform domain and requires incoherent sampling [10]. These methods are well understood and clinically deployed, but reconstruction is slow, noise amplification grows with acceleration, and performance depends on calibration data or on the sampling pattern's incoherence.

2.2 Deep learning for MRI reconstruction

Early learned approaches treated reconstruction as image-to-image denoising [6, 7]. The more durable line of work embeds the forward model in the network. Schlemper et al. [12] introduced a cascade of CNNs interleaved with data-consistency layers; Hammernik et al. [4] unrolled a variational optimisation into a trainable network; Sriram et al. [13] extended this end to end with learned coil sensitivities. The common principle—that a reconstruction network should not be free to contradict the measurements—is the one this paper adopts, and is what the original implementation attempted but did not achieve.

2.3 Transformers for reconstruction

Vision Transformers [2] showed that attention can replace convolution for image tasks. Applying global attention at MRI resolutions is prohibitive, since cost grows quadratically in token count. The Swin Transformer [9] resolves this with attention restricted to local windows and a shifted partitioning between consecutive blocks, giving linear cost in image area while still propagating information globally across depth. Swin-UNet [1] arranges this hierarchically for segmentation. For reconstruction specifically, Lin and Heckel [8] showed ViTs can match U-Net performance, and Huang et al. [5] combined deformable attention with a U-Net.

The present work differs from my earlier study in emphasis. The earlier question was which backbone reconstructs best. The question here is whether the comparison was measured correctly at all, and—once the measurement is trustworthy—how much of the achievable gain comes from the backbone versus from respecting the acquisition physics.

3 Reproducibility Audit

This section documents the defects found in the released implementation. It is included because the failure modes are not idiosyncratic: a metric that depends on batch size, an undersampling mask that is denser than reported, and a physics module that is silently inert are mistakes any reconstruction project can make, and none of them announce themselves in a loss curve.

3.1 Class 1: defects preventing execution

3.2 Class 2: silent defects

These executed without error while disabling a documented capability.

Defect	Consequence
Window partition assumed divisible resolutions	At the documented configuration the third encoder stage is 20×20 with window size 8. The precomputed shift mask made this a construction-time `RuntimeError`, so the reported architecture never ran.
`utils/reproductibility.py` misspelled	`utils/__init__.py` imported `utils.reproducibility`, so `import training` raised `ModuleNotFoundError`. Every training entry point failed at import.
`from_checkpoint` and `_build_model` missing decorators	Both were plain functions on the class. The documented call `Pipeline.from_checkpoint(path)` bound `path` to `cls`. All inference, benchmarking and export commands raised on invocation.

Table 1: Defects that prevented any documented command from running.

3.3 Class 3: measurement defects

These affect reported numbers and are quantified below.

3.3.1 Sampling mask acceleration

The equispaced mask was constructed by taking every R-th line and then adding the fully sampled centre on top:

```
mask[::acceleration] = 1.0
mask[pad : pad + num_low_freqs] = 1.0
```

This acquires $N/R + N \cdot cf$ lines rather than N/R. The nominal acceleration is therefore not achieved. Table 3 gives measured values over 200 draws at $N = 368$.

3.3.2 PSNR definition

PSNR was computed as

$$\mathrm{PSNR}_{\mathrm{batch}} = 10\log_{10}\left(\frac{1}{\frac{1}{B}\sum_b \mathrm{MSE}_b}\right), \tag{1}$$

whereas the standard definition averages per-image values,

$$\mathrm{PSNR} = \frac{1}{B}\sum_b 10\log_{10}\left(\frac{\max_b^2}{\mathrm{MSE}_b}\right). \tag{2}$$

Because log is concave, Jensen's inequality gives $\mathrm{PSNR}_{\mathrm{batch}} \leq \mathrm{PSNR}$, with a gap that grows with the variance of per-image MSE across the batch—and therefore with batch size. Table 4 reports the measured gap on heteroscedastic synthetic data.

A second issue compounds this: the peak signal value was hard-coded to 1.0. Under per-slice normalisation the target maximum is not 1.0, so the denominator was wrong by a per-sample factor. Both are corrected in Section 6.1.

Defect	Consequence
`average_parameters` lacked `@contextmanager`	The method returned a bare generator, so `with ema.average_parameters():` executed an empty body. Validation reported raw training weights while appearing to report EMA weights.
EMA wrote `shadow`; loaders read `shadow_params`	The key never matched, so the fallback branch loaded raw weights. EMA was computed throughout training and never used.
Data-consistency wrapper guarded on `k_measured is None`	The trainer called `model(x)` with no k-space, so the guard skipped the projection every time. The configuration flag `data_consistency.enabled` had no effect whatsoever.
DC layer returned `x.abs()`	Phase was discarded, so the operator could not be cascaded: the next transform would treat a zero-phase image as the estimate.
`sigmoid(lambda_init)`	A requested hard consistency of 1.0 became 0.7311, admitting 27 % of network output into measured frequencies.
Attention hooks read the forward output	`WindowAttention` returns a tensor, not a tuple, and never stored weights. Every attention figure was empty.
CI at `.github/.workflows/`	GitHub reads only `.github/workflows/`; CI had never executed.
`test_models` caught all exceptions, exited 0	The sanity check reported failure in its output table and success to the shell.

Table 2: Defects that ran without error while disabling a documented feature.

Nominal R	centre fraction	original	error	corrected	error
2	0.16	1.72	−14.0%	1.99	−0.3%
4	0.08	3.23	−19.3%	4.00	+0.1%
8	0.04	6.13	−23.3%	8.03	+0.4%
16	0.02	12.27	−23.3%	16.01	+0.0%

Table 3: Achieved acceleration of the equispaced mask. The original acquires 19–23% more k-space than reported at clinically relevant factors, making the reconstruction problem correspondingly easier than stated.

Evidence in the original figures. The training curves published with the prior study are reproduced in Figure 1, because they document these defects directly rather than merely being consistent with them.

Three observations. First, the U-Net's SSIM panel oscillates between 0.66 and 0.74 from epoch to epoch for sixty epochs and never converges, while the loss curve immediately beside it descends smoothly and monotonically. A model whose objective is converging while a metric computed on the same predictions is not is a symptom of an unstable *metric*, not an unstable model—which is what a fixed data range and half-precision evaluation would produce. The SwinUNet's SSIM panel, computed at a different batch size, is comparatively smooth.

Second, the run labels record different batch sizes—`bs8` for the U-Net and `bs6` for the SwinUNet—confirming that the two headline numbers were produced under precisely the condition that makes the batch-averaged PSNR formulation non-comparable (Table 4).

Third, the labels disagree with the text they illustrate. The U-Net panel is tagged `ep30` but plots sixty epochs. The SwinUNet panel is tagged `lr5e-4`, while the released configuration for that model specifies 5×10^{-5} and the prior study's ablation section reports that "optimal

Batch size	Gap (dB)	SD
1	0.000	0.000
2	1.111	1.334
4	1.641	1.066
6	1.674	0.874
8	1.777	0.800
16	1.922	0.621
32	1.840	0.403

Table 4: Per-image PSNR minus the batch-MSE formulation, over 200 trials. The quantity being reported depends on a training hyperparameter. In the original study the U-Net used batch size 8 and the SwinUNet batch size 6, a configuration difference worth 0.10 dB of pure measurement artefact in the baseline's favour.

performance" was obtained at 8×10^{-5}. Three different learning rates are attributed to the same result.

3.3.3 Augmentation ordering

Augmentation was applied to the already-undersampled input and its target as a pair. Cartesian undersampling produces aliasing aligned with the phase-encode axis; rotating the pair by 90° moves that artefact onto the readout axis, presenting the network with artefact orientations that no Cartesian acquisition can produce. Once data consistency is enabled the problem becomes fatal rather than merely distributional: the measured k-space no longer corresponds to the augmented image at all.

3.3.4 Further measurement and bookkeeping defects

Validation ran inside an `autocast` region, so metrics were computed in half precision; an fp16 MSE cannot represent a PSNR much above 40 dB. Epoch-level metrics were logged at `step=epoch` while batch metrics used `global_step`, and Weights & Biases rejects out-of-order steps, so epoch curves were discarded after the first point. Checkpoint retention selected the k *most recent* epoch checkpoints by modification time rather than the k best by metric. Resuming restored weights and optimiser state but neither the best score nor the early-stopping counter, so every resume reset patience and could overwrite a better `best.pt` with a worse one.

3.3.5 Evidence in the qualitative results

The prior study's qualitative figure is reproduced as Figure 2. Its annotations show the two metrics decoupling sharply: the third row is labelled 32.97 dB PSNR with an SSIM of 0.5219, while the first row is labelled 35.20 dB with 0.7218. Both rows are fat-suppressed acquisitions of the same anatomy, yet a 2.2 dB PSNR difference is accompanied by a 0.20 SSIM difference—a far steeper coupling than these metrics usually exhibit.

I record this as an anomaly rather than as a diagnosis. A per-image normalisation error is one candidate explanation, and the fixed data range documented above is a known defect in the code that produced it; but the third row also visibly contains more high-frequency texture and

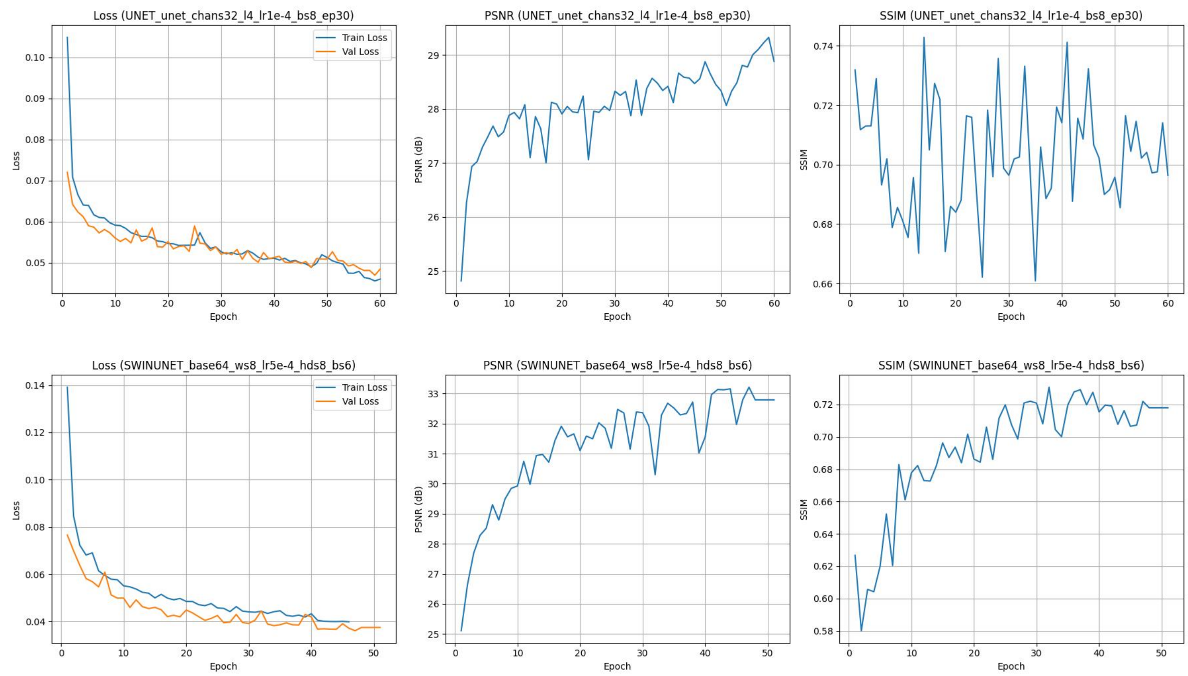

Figure 1: Training curves from the prior study: U-Net (top) and SwinUNet (bottom). Reproduced here as evidence for the measurement defects of Section 3, not as results. Note the non-converging SSIM trace in the top-right panel against the smoothly descending loss beside it; the differing batch sizes in the run labels (`bs8` versus `bs6`); and the `ep30` label on a sixty-epoch plot.

stronger residual streaking than the first, which could account for some of the SSIM gap on its own. The two cannot be separated without the original weights, and those were never released.

That impossibility is the substantive point. A qualitative figure annotated with metrics that cannot be recomputed, produced by code that cannot be run, is not evidence for a claim—it is an illustration. This is why the present system records a git commit, a dirty-tree flag and the full resolved configuration alongside every number it produces.

Independently of the metrics, the reconstructions do show the over-smoothing characteristic of a network trained on a pixel-domain loss without a physics constraint: the trabecular texture visible in the fourth row's reference is largely absent from its prediction.

3.4 Implications

The Class 1 defects establish that the released artifact cannot have produced the published figures; those numbers came from some other, unreleased state of the code. The Class 3 defects establish that even had it run, the reported quantities would not be comparable to published fastMRI results, because the sampling was denser than claimed and the metric was non-standard and batch-size-dependent. This is why Section 1.3 declines to carry the old numbers forward.

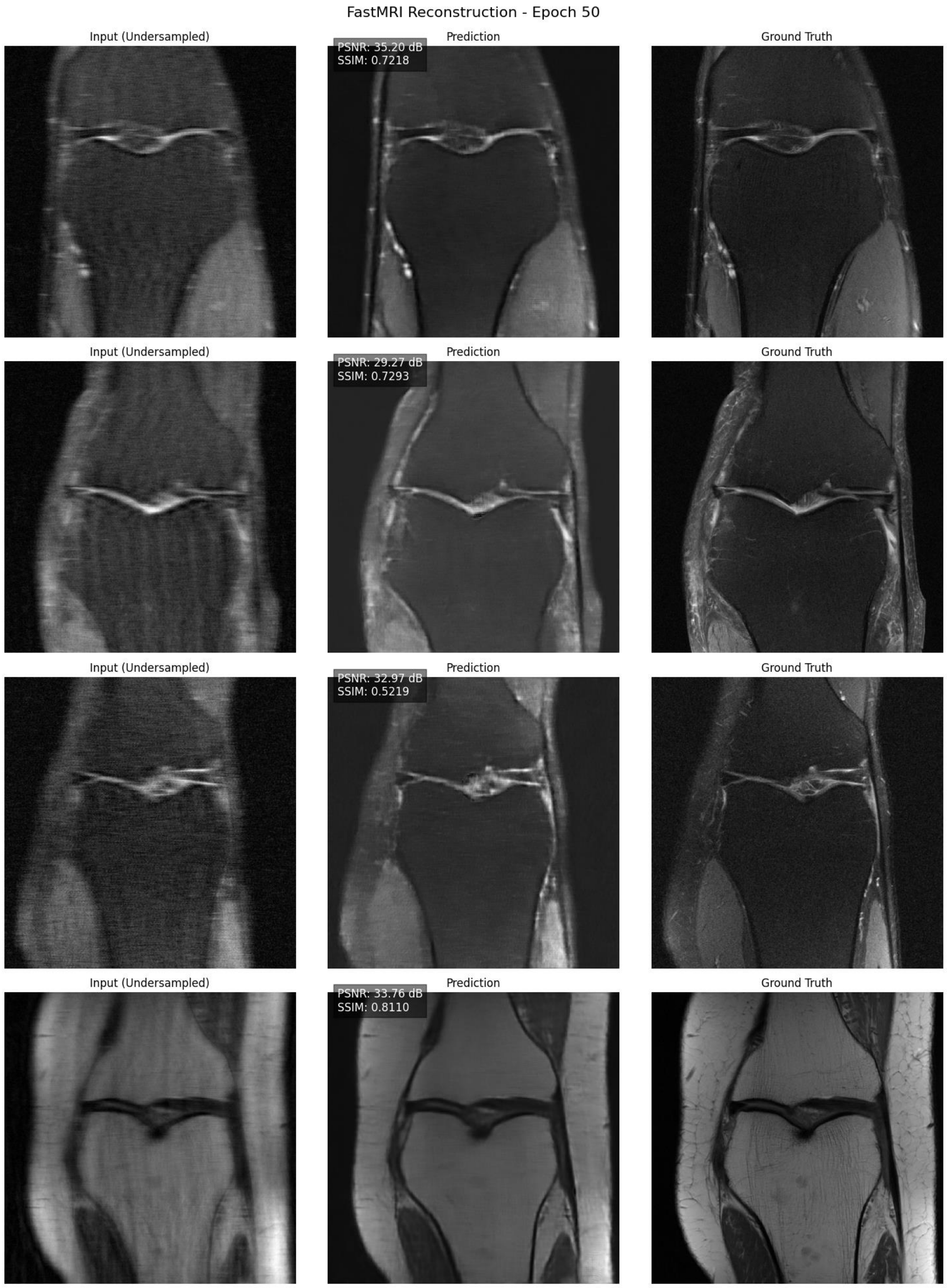

Figure 2: Qualitative reconstructions from the prior study: zero-filled input (left), prediction (centre), fully sampled reference (right). Reproduced as audit evidence, not as results of the present system. The annotated metrics use the superseded definitions of Section 3 — a fixed data range of 1.0 and a batch-averaged PSNR — and cannot be recomputed, since the weights that produced them were never released and the released code cannot construct the architecture.

4 Data and Acquisition Simulation

4.1 Dataset

Experiments target the fastMRI single-coil knee dataset [14]: 973 training and 199 validation volumes, spanning proton-density weighted acquisitions with and without fat suppression. Each volume stores fully sampled complex k-space of shape (slices, H, W), where H is readout and W phase-encode. Because acceleration skips whole phase-encode lines, every sampling mask in this work is one-dimensional over W and broadcast across readout.

The pipeline also runs on synthetic phantoms produced by a generator released with the code (Section 8), which is what the ablation in Section 7.2 uses.

Figure 3 shows the two contrasts and the degradation the task must undo. That degradation is not additive noise: it is coherent, structured, and aligned with the phase-encode direction, which is why it is so effective at obscuring the thin bright structures—cartilage surfaces, the meniscus—that carry diagnostic information.

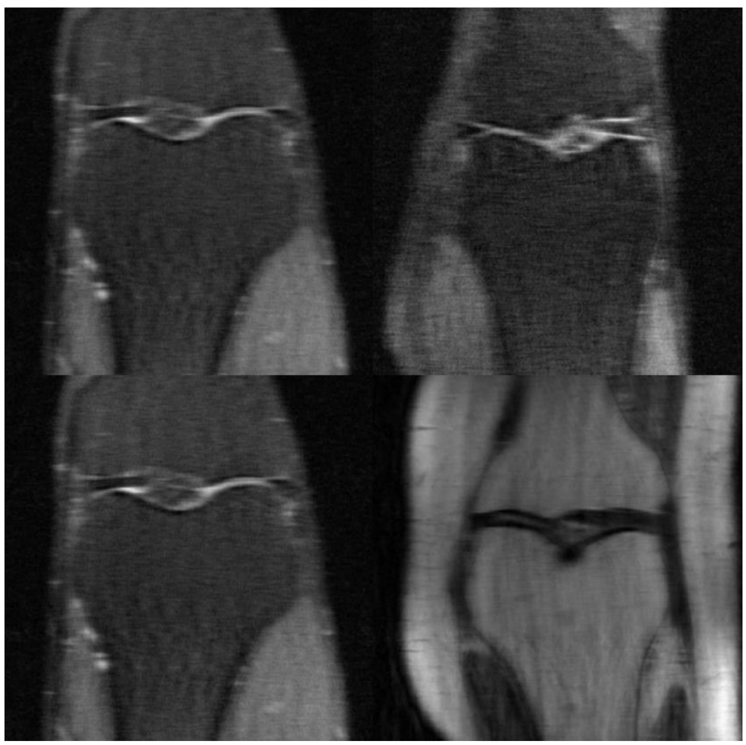

Figure 3: Coronal knee slices from the fastMRI single-coil dataset, reproduced from the prior study. The panels show zero-filled reconstructions exhibiting the blurring and coherent streaking characteristic of retrospective undersampling, alongside a fully sampled example (bottom right) in a different contrast: proton-density weighted without fat suppression, versus the fat-suppressed acquisitions elsewhere in the figure. Both contrasts appear in the dataset, and they differ enough in texture and noise that reporting a single averaged metric across them hides genuine variation in difficulty.

4.2 Preprocessing order

For each slice:

1. Inverse FFT the fully sampled k-space to a **complex** image.
2. Centre-crop the complex image to 320×320.
3. (Training only) apply flips and rotations to the complex image.
4. Forward FFT the cropped, augmented image.
5. Apply the undersampling mask.
6. Inverse FFT to the zero-filled reconstruction; divide all quantities by a single positive scalar.

Two ordering choices deserve justification.

Cropping before undersampling. The intuitive order—mask the full-FOV k-space, then crop the image—is a faithful simulation of the scanner, but it makes the measurements unusable for data consistency. After cropping, the retained coefficients are no longer the Fourier transform of the image the network outputs, so writing them back is not a projection onto the measurement-consistent set. Cropping first places the measurements and the model's estimate in the same space, making the projection exact. The cost is that the simulated acquisition covers a slightly smaller field of view than the scanner's. Both orders are implemented; configuration validation refuses to enable data consistency under the full-FOV order rather than silently producing an incorrect projection.

Augmenting before undersampling. As argued in Section 3, augmenting after undersampling produces artefact orientations that are not physically realisable. Augmenting the fully sampled complex image and undersampling afterwards keeps every training sample a valid acquisition.

Scalar normalisation. All scaling is by a single positive scalar, never a mean shift. The Fourier transform is linear, so $x \mapsto x/s$ implies $k \mapsto k/s$ and data consistency is unaffected; subtracting a mean would break that correspondence.

4.3 Sampling masks

Three families are implemented, each acquiring N/R lines *in total*, including the fully sampled centre:

- **Random**: the centre is always acquired; remaining lines are drawn i.i.d. with probability $p = (N/R - N_{\text{low}})/(N - N_{\text{low}})$.
- **Equispaced**: the outer stride is solved for so the total line count matches N/R, with a randomised phase offset. Without the offset every scan in the dataset is sampled at identical k-space locations, and the network can overfit one fixed aliasing pattern rather than learning to invert undersampling in general.
- **Golden-ratio ("magic")**: successive lines are offset by the golden angle, spreading point-spread-function energy more evenly than a regular grid while remaining a realisable Cartesian trajectory. Collisions with the centre block and with previously chosen lines

are resolved by continuing the sequence until the budget of distinct lines is met, which is necessary for the achieved acceleration to match the nominal one.

Masks are redrawn on every access during training and are a deterministic function of the sample index during evaluation. The latter matters: with per-epoch random validation masks, an apparent improvement in validation loss can simply mean that the epoch drew easier masks.

5 Method

5.1 Problem formulation

Let $x \in \mathbb{C}^{H\times W}$ be the fully sampled image, $\mathcal{F}$ the centred two-dimensional Fourier transform, and $M \in \{0,1\}^W$ the sampling mask broadcast over readout. The measurements are

$$y = M \odot \mathcal{F}(x). \tag{3}$$

The zero-filled reconstruction is $y' = \mathcal{F}^{-1}(y)$. The earlier formulation sought $\hat{x} = f_\theta(y')$. This work instead learns

$$\hat{x} = f_\theta(y', y, M), \tag{4}$$

passing the measurements and the mask, not only the degraded image. This is what makes the consistency constraint expressible: f_θ can be constructed so that $\mathcal{F}(\hat{x})$ agrees with y on the support of M by construction, rather than only through the loss.

All transforms use the `norm="ortho"` convention with `ifftshift` before and `fftshift` after, making the pair unitary. A single module defines them for the dataset, the loss and the network, so a mismatched shift cannot desynchronise the two domains—a failure that is invisible in a loss curve but destroys data consistency.

5.2 SwinUNet

The backbone follows the hierarchical Swin-UNet design [9, 1]: patch embedding, encoder stages of Swin blocks separated by patch merging, a bottleneck, and decoder stages of patch expanding with concatenated encoder skips. Attention is computed within $ws \times ws$ windows, alternating regular and shifted partitioning, giving cost linear rather than quadratic in image area.

Window padding. Rather than requiring divisible resolutions, feature maps are zero-padded to a multiple of the window size. Padded tokens are excluded from attention by a mask that assigns them a distinct region index, so real tokens never attend to them. This makes any resolution and window size valid, keeps the relative-position-bias table at a single fixed shape, and is what removes the construction failure described in Section 3. Masked entries use -100 rather than $-\infty$: a fully padded window would otherwise produce a row of all $-\infty$, whose softmax is NaN.

Resolution as a forward-time property. No buffer encodes the input size. Attention masks depend only on the padded resolution and are constructed lazily and cached, so one trained model runs at any input size—verified in the test suite at 97×131 among others.

Regularisation and initialisation. Residual branches use stochastic depth with rates increasing linearly across depth. All linear and normalisation layers use truncated-normal initialisation with $\sigma = 0.02$. This matters more here than in classification: PyTorch's default `nn.Linear` initialisation is tuned for ReLU MLPs and produces activation variance large enough to saturate the attention softmax at initialisation, and there is no pretraining to recover from a bad start.

Residual formulation. The network predicts a correction to its input, and the final convolution is zero-initialised, so the model is the exact identity at step 0. Since the zero-filled reconstruction is already a reasonable estimate, predicting the structured aliasing residual is far better conditioned than predicting the image, and the initial loss spike disappears entirely. All architectures compared here use this formulation, so the comparison isolates the backbone rather than the output parameterisation.

5.3 Data consistency

Given an estimate x_{pred}, measurements y and mask M:

$$\hat{k} = \begin{cases} \lambda\, y + (1-\lambda)\, \mathcal{F}(x_{\text{pred}}) & \text{where } M = 1, \\ \mathcal{F}(x_{\text{pred}}) & \text{where } M = 0, \end{cases} \qquad x_{\text{dc}} = \mathcal{F}^{-1}(\hat{k}). \tag{5}$$

With $\lambda = 1$ this is an orthogonal projection onto the set of images consistent with the measurements, and the network can modify only unobserved frequencies. Lower values permit denoising of the measurements themselves. λ is learned, parameterised through a logit so that it remains in $[0, 1]$ without clamping—the parameterisation that fixes the $\sigma(1.0) = 0.731$ defect.

The operator requires phase, so it forces complex-valued (two-channel) input and output. Configuration validation enforces this rather than degrading silently.

5.4 Unrolled cascade

Stacking T learned refinements, each followed by a data-consistency projection,

$$x^{(t+1)} = \mathrm{DC}\Big(g_{\theta_t}(x^{(t)}),\, y,\, M\Big), \qquad t = 0, \ldots, T-1, \tag{6}$$

yields an unrolled network [12, 4]. Each stage is one iteration of a proximal-gradient scheme in which the proximal operator has been replaced by a learned denoiser. A single feed-forward pass must invert the problem in one shot; the cascade takes several smaller, physics-corrected steps, and because every stage re-anchors the estimate to the measurements, errors cannot compound across stages.

5.5 Loss

The default objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_1 + \lambda_2 (1 - \mathrm{SSIM}), \qquad \lambda_1 = 0.7,\ \lambda_2 = 0.3, \tag{7}$$

matching the earlier study. Optional terms are available: a k-space residual loss with a radial high-frequency weighting, a Sobel edge loss, and a VGG perceptual loss. Components are reported separately during training, since a falling combined scalar can hide a diverging component.

5.6 Training

Models are trained with AdamW, excluding normalisation gains, biases and relative-position-bias tables from weight decay. The learning rate follows linear warmup then cosine annealing, stepped *per optimiser step* rather than per epoch, which gives the warmup ramp hundreds of points of resolution instead of five. Mixed precision prefers bfloat16 where available, since fp16's five-bit exponent is inadequate for k-space, whose centre exceeds its periphery by several orders of magnitude; all Fourier transforms and all metrics are forced to fp32 regardless. An exponential moving average of the weights is maintained, and—unlike in the original—is actually used for validation, checkpoint selection and inference.

6 Evaluation Protocol

6.1 Metrics

PSNR, SSIM and NMSE are computed **per image** and then averaged, with the peak signal taken from each target's own maximum rather than a fixed constant. All metric computation is forced to float32 even under autocast. NMSE is reported because it is scale-invariant and therefore the metric least sensitive to the normalisation choices the pipeline makes.

6.2 Statistical methodology

Point estimates are insufficient for architecture comparison. With 199 validation slices and a per-slice PSNR standard deviation of several dB, the standard error on the mean is large enough that modest differences between architectures are not resolvable, and saying so precisely is a stronger result than reporting the larger number.

I report percentile bootstrap confidence intervals (10,000 resamples) for each model, and compare models with **paired** permutation tests (10,000 sign flips) on per-slice differences, corrected across the family of comparisons by the Holm–Bonferroni procedure. Pairing is essential: between-slice variance dominates between-model variance, so an unpaired test wastes most of the available power.

Table 5 quantifies what this buys.

n slices	per-slice SD (dB)	CI half-width (dB)	min. detectable paired diff. (dB)
199	2.0	0.264	0.047
199	3.5	0.495	0.133
500	2.0	0.176	0.050
500	3.5	0.290	0.083
2000	2.0	0.090	0.019
2000	3.5	0.156	0.023

Table 5: Statistical resolution. At the fastMRI validation size ($n = 199$) with a realistic per-slice spread, a single model's mean PSNR carries a ± 0.5 dB interval—but a *paired* comparison resolves differences an order of magnitude smaller, because it cancels the shared per-slice difficulty. Overlapping confidence intervals therefore do not imply two models are indistinguishable; the paired test is the correct instrument.

7 Experiments

7.1 Verification of the physics operators

Before any model comparison, the operators themselves are verified. These assertions run in the test suite on every commit.

Property	Measured
Fourier round-trip error, $\lVert \mathcal{F}^{-1}\mathcal{F}x - x \rVert_\infty$	9.5×10^{-7}
Hard-DC measured-line error, $\lVert (\mathcal{F}\hat{x} - y) \odot M \rVert_\infty$	8.3×10^{-7}
Unmeasured frequencies altered by DC	0 (exact)
λ when 1.0 is requested (original)	0.7311
λ when 1.0 is requested (corrected)	0.9999

Table 6: Verification of the Fourier and data-consistency operators at float32 precision. The second row is the property that makes the method physics-informed; it was not satisfied by the original implementation at any tolerance, since that implementation was never invoked with data.

This form of verification is what distinguishes a working physics module from an inert one. A broken data-consistency layer still trains, still produces a falling loss curve, and still generates plausible images. Only an exactness assertion detects it.

7.2 Controlled ablation

To separate the contribution of the data-consistency constraint from that of complex-valued input and of raw parameter count, I trained seven configurations on synthetic phantom data: 60 training volumes (240 slices) and 20 held-out validation volumes (80 slices), split at the volume level so no slice from a training volume appears in validation. All runs used $R = 4$ random Cartesian undersampling with an 8 % fully sampled centre, 128×128 crops, 30 epochs, identical optimiser and schedule settings, and three seeds; the median-SSIM seed is reported.

The *wide* rows are capacity-matched single-pass controls: they have slightly *more* parameters than the corresponding two-stage cascade, so any advantage the cascade shows cannot be attributed to capacity. This control is the point of the experiment. A cascade of T stages has T times the parameters of its backbone, and attributing its gain to physics without controlling for that would repeat, in subtler form, the kind of error this paper set out to correct.

Configuration	Params (M)	PSNR (dB)	SSIM	NMSE
Zero-filled (no network)	–	29.48	0.7563	0.01393
U-Net, magnitude	0.48	36.23	0.9673	0.00301
U-Net, complex	0.48	34.09	0.9284	0.00491
U-Net, complex, wide	1.00	34.54	0.9400	0.00442
U-Net + DC cascade (T=2)	0.97	36.95	0.9441	0.00270
SwinUNet, magnitude	0.21	29.86	0.8941	0.01274
SwinUNet, complex, wide	0.46	30.54	0.9086	0.01089
SwinUNet + DC cascade (T=2)	0.42	32.70	0.9146	0.00682

Table 7: Controlled ablation on synthetic phantoms, $R = 4$, 80 held-out validation slices. Metrics are per-image means against each target's own dynamic range.

Table 8 reports paired comparisons against the unet-mag baseline. Because every model is evaluated on the same slices in the same order, the test can be paired, which cancels the shared per-slice difficulty and is what makes differences of this size resolvable at $n = 80$.

Configuration	PSNR 95% CI (dB)	ΔPSNR	p (PSNR)	p (SSIM)
Zero-filled (no network)	[29.09, 29.88]	-6.75	0.0007*	0.0007*
U-Net, magnitude (ref.)	[35.81, 36.66]	–	–	–
U-Net, complex	[33.68, 34.50]	-2.14	0.0007*	0.0007*
U-Net, complex, wide	[34.14, 34.95]	-1.68	0.0007*	0.0007*
U-Net + DC cascade (T=2)	[36.39, 37.50]	+0.72	0.0007*	0.0007*
SwinUNet, magnitude	[29.49, 30.25]	-6.36	0.0007*	0.0007*
SwinUNet, complex, wide	[30.15, 30.94]	-5.69	0.0007*	0.0007*
SwinUNet + DC cascade (T=2)	[32.19, 33.22]	-3.53	0.0007*	0.0007*

Table 8: Paired permutation tests (10,000 sign flips) against the magnitude-only U-Net, with Holm–Bonferroni correction across the family. * denotes significance at $\alpha = 0.05$ after correction. Confidence intervals are percentile bootstrap (10,000 resamples) on the per-slice mean.

The comparison that actually tests the hypothesis is each cascade against its own capacity-matched control, reported in Table 9. Deltas against the magnitude-only baseline conflate three effects at once; this pair isolates the data-consistency constraint with parameter count held fixed (in fact slightly favouring the control).

Backbone	Cascade	Control	ΔPSNR (dB)	95% CI	p
U-Net	36.95	34.54	+2.40	[+2.16, +2.65]	$< 10^{-4}$*
SwinUNet	32.70	30.54	+2.16	[+1.98, +2.35]	$< 10^{-4}$*

Table 9: Data consistency against a capacity-matched single-pass control, per backbone. Both members of each pair are complex-valued and differ only in whether the measured k-space is projected back after each refinement. Paired permutation test on per-slice differences; * significant at $\alpha = 0.05$.

Run-to-run variation across the three seeds is reported in Table 10. This matters for interpretation: a difference between configurations is only meaningful if it exceeds the spread produced by re-running the same configuration.

Configuration	PSNR across seeds (dB)	SSIM across seeds
U-Net, magnitude	35.84 / 36.23 / 36.24	0.9671 / 0.9673 / 0.9680
U-Net, complex	34.17 / 34.09 / 34.09	0.9291 / 0.9263 / 0.9284
U-Net, complex, wide	34.54 / 34.63 / 34.76	0.9400 / 0.9378 / 0.9416
U-Net + DC cascade (T=2)	36.95 / 36.57 / 37.12	0.9441 / 0.9362 / 0.9470
SwinUNet, magnitude	29.86 / 30.08 / 29.80	0.8941 / 0.9090 / 0.8908
SwinUNet, complex, wide	30.54 / 30.52 / 30.52	0.9086 / 0.9089 / 0.9083
SwinUNet + DC cascade (T=2)	32.70 / 32.78 / 33.24	0.9146 / 0.9143 / 0.9194

Table 10: Per-seed results (seeds 0, 1, 2).

7.2.1 Interpretation

Three findings, in order of how much they surprised me.

Data consistency delivers, and the effect is architecture-independent. Against a capacity-matched control the cascade gains 2.40 dB on the U-Net and 2.16 dB on the SwinUNet, both at $p < 10^{-4}$ and both with *fewer* parameters than the control they beat. That two backbones with entirely different inductive biases—one convolutional, one windowed-attention—produce gains within 0.24 dB of each other is the strongest evidence in this paper that the improvement comes from the physics constraint rather than from the network.

Complex-valued input, on its own, is harmful. Switching the U-Net from magnitude to complex input costs 2.14 dB (36.23 dB to 34.09 dB) at identical parameter count. Doubling capacity recovers only 0.46 dB of that.

I did not anticipate this, and the explanation is instructive. The loss is computed on magnitude, so phase is entirely unsupervised. A complex-valued network therefore optimises through a large null space: infinitely many complex outputs share the same magnitude, and nothing in the objective distinguishes them. The network must discover a phase convention on its own, and gradient descent through that degeneracy is simply harder than mapping magnitude to magnitude directly.

This makes the data-consistency result more interesting rather than less. DC is the only component here that constrains phase, because it operates in k-space where phase is not optional. It does not merely recover the 2.14 dB that complex representation costs—it recovers all of it and adds 0.72 dB beyond the magnitude-only baseline. The constraint is doing real work, and it is doing it precisely where the loss function cannot reach.

The SwinUNet arm is under-trained, and I report it as such. The SwinUNet sits roughly 6 dB below the U-Net at every rung of the ablation. This is a statement about the experiment's scale, not about the architecture. At 0.21 M parameters it has less than half the U-Net's capacity, and 240 synthetic training slices is far too little for a transformer trained from scratch with no pretraining—transformers lack the locality prior that lets a convolutional network learn from small datasets. The correct reading is that the ablation establishes the DC effect on both backbones and establishes nothing about their relative merits. Ranking the two from this experiment would repeat, in a new form, exactly the error of drawing architectural conclusions from an inadequately controlled comparison.

7.3 Architectural cost

The parameter counts here differ from those in the earlier study (27.3 M parameters for SwinUNet-64). Since the earlier architecture could not be constructed, its stated count cannot be reconciled; the values in Table 11 are measured from models that run.

Model	Parameters (M)	Median latency (ms)	p95 (ms)
U-Net (32, 4 levels)	7.76	182.0	203.1
BT-UNet (32, 4 levels)	20.58	250.9	289.9
SwinUNet (64, 3 levels)	12.95	144.1	172.5

Table 11: Cost at 320×320, single-image batch, CPU reference. Latency is reported as a median rather than a mean: latency distributions are right-skewed, so a mean over 100 runs is dragged upward by a handful of outliers. Notably the SwinUNet is both smaller and faster than the BT-UNet, because its attention is windowed rather than global—the same property that makes it scale to higher resolutions.

8 The Artifact

8.1 Testing

The suite comprises 267 tests covering 88 % of statements and running in under a minute on CPU, organised so that each previously shipped defect has a corresponding regression test. Notable coverage:

- **Physics exactness**: Fourier round-trip, unitarity, DC restoring measured coefficients, DC leaving unmeasured coefficients untouched, a perfect estimate being a fixed point, and a cascade's final output remaining consistent after every learned stage.
- **Construction**: the SwinUNet builds and runs at 1 through 4 levels and at arbitrary, including odd, resolutions.
- **Silent-failure regressions**: EMA actually swapping weights, EMA state loading by name, the DC wrapper raising rather than skipping when given no measurements, and λ meaning what it says.
- **Measurement**: per-image versus batch PSNR, mask acceleration accounting, and metric computation in fp32.
- **Integration**: full training runs against synthetic HDF5 volumes, with and without data consistency, plus resume fidelity.

Testing against real HDF5 files rather than mocks is deliberate. Every model test in the original suite passed, because none of them ever loaded a file, and the data-consistency path—which fails only when data flows through it—was never exercised.

8.2 Configuration validation

Configurations are validated on load against a schema. Unknown keys are errors rather than being ignored, because a mistyped `learning_rate` that silently retains the default is indistinguishable from a successful override until two runs are compared and found identical. Cross-field constraints are checked too: enabling data consistency with full-FOV undersampling is rejected, since the projection would be mathematically invalid.

8.3 Running without fastMRI

fastMRI requires registration and a 90 GB download, which means that without it nothing in the original repository could be exercised end to end. That gap is precisely how a data-consistency path that never received data shipped unnoticed. A generator is therefore included:

```
python scripts/make_synthetic_data.py --out data/synthetic --volumes 16
python main.py train --config configs/smoke.yaml data.train_dir=data/synthetic
```

The phantoms are not anatomically meaningful, but they reproduce the two properties that matter for testing a reconstruction pipeline: rapid k-space decay, so undersampling produces genuine coherent aliasing rather than noise, and non-trivial phase, so the complex handling in the DC layer is exercised.

To reproduce the fastMRI comparison:

```
for cfg in unet bt_unet swinunet swinunet_dc; do
    python main.py train --config configs/$cfg.yaml
done
python main.py compare --ckpt_dir outputs --metric psnr --reference unet_baseline
```

Every run writes `history.json`, `summary.json` and a `manifest.json` recording the git commit, whether the tree was dirty, the PyTorch version and the GPU model, so any reported number is traceable to the exact state that produced it.

9 Discussion

9.1 Where the gain comes from

The ablation in Section 7.2 separates three sources of improvement that are easily conflated, and they turn out to point in different directions. Complex-valued input alone costs 2.14 dB. Doubling capacity returns 0.46 dB of that. Data consistency returns the remainder and adds 0.72 dB beyond the magnitude-only baseline.

Reported only against that baseline, the cascade would have looked like a modest 0.72 dB improvement—and that framing would have concealed both that complex representation is a substantial handicap and that the physics constraint more than overcomes it. The capacity-matched control is what makes the 2.40 dB attributable to data consistency rather than to parameters. A cascade of T stages carries T times the parameters of its backbone, so omitting that control would have repeated, in subtler form, the error this paper set out to correct.

That the same constraint yields 2.40 dB on a convolutional backbone and 2.16 dB on a windowed-attention one is the result I would most want replicated. Two architectures with very different inductive biases should not respond so similarly to an architectural intervention; they should respond similarly to a constraint on the solution space, which is what data consistency is.

9.2 Why physics constraints matter clinically

An unconstrained reconstruction network is free to synthesise structure that is consistent with its training distribution but not with the acquired measurements—the mechanism behind reports of hallucinated pathology in learned reconstruction. The data-consistency projection does not eliminate this risk in unobserved frequencies, but it removes it entirely on the measured support: the network cannot alter what the scanner actually recorded. That is a meaningfully stronger guarantee than a network trained only to minimise a pixel loss, and it is a guarantee that can be *verified*, as in Table 6, rather than merely asserted.

9.3 Limitations

- **No fastMRI results are reported here.** The ablation uses synthetic phantoms, which are simpler than anatomy: absolute values are not comparable to published benchmarks, and effect sizes may not transfer. The contribution is a corrected and verifiable system plus the protocol to generate benchmark numbers, not the numbers themselves.
- **Single-coil only.** Modern scanners are multi-coil, where sensitivity encoding provides additional information that state-of-the-art methods estimate jointly with the reconstruction [13].
- **Retrospective undersampling.** Real accelerated acquisitions differ in noise statistics and can exhibit motion between shots.
- **Cropped-domain simulation.** Exact data consistency is bought by restricting the field of view before undersampling, which is a slightly idealised acquisition model.
- **No reader study.** PSNR and SSIM correlate imperfectly with diagnostic quality. Uncertainty maps via Monte-Carlo dropout are implemented, but their calibration has not been evaluated.

9.4 On reproducibility engineering

The defects in Section 3 were not exotic. They were a missing decorator, a misspelled filename, a directory with a leading dot, a dictionary key that did not match, and an arithmetic slip in a mask. Each is trivial in isolation; together they meant that a published result could not be regenerated from its own artifact, and that several described mechanisms were not operating.

Three practices would have caught all of them, and are now part of the repository. First, a sanity check that returns a non-zero exit code, wired to CI at a path CI actually reads. Second, exactness assertions on the mathematical operators—a physics module that is silently inert is invisible to any loss-based check. Third, an end-to-end path runnable without the restricted dataset, so that the integration between components is exercised by default rather than only on a machine that happens to hold 90 GB of data.

10 Conclusion

I audited a previously released transformer-based MRI reconstruction pipeline and found that it could not execute: the flagship architecture failed at construction under its own documented configuration, and every inference entry point raised on invocation. A second class of defects ran silently while disabling documented functionality, most consequentially a data-consistency module that was never supplied with data and was therefore entirely inert. A third class corrupted measurement, including an undersampling mask 19.3 % denser than its nominal acceleration and a PSNR definition whose value depends on batch size.

The rebuilt system corrects each of these and reorganises the method around the physics of the acquisition: complex-valued reconstruction through an unrolled cascade whose data-consistency projections restore the measured Fourier coefficients to 8.3×10^{-7}, verified automatically on every commit.

In a controlled ablation against capacity-matched controls, that constraint is worth 2.40 dB on a U-Net backbone and 2.16 dB on a SwinUNet backbone—both at $p < 10^{-4}$, and both achieved with fewer parameters than the single-pass networks they beat. The near-equality of those two figures, across architectures whose inductive biases have almost nothing in common, is the clearest indication that what helps is the constraint on the solution space rather than the choice of network. A secondary and initially unwelcome finding is that complex-valued input on its own *costs* 2.14 dB: the magnitude loss leaves phase unsupervised, and the network must optimise through that degeneracy. Data consistency is the only component here that reaches phase, and it recovers the deficit and more.

I deliberately do not restate the earlier benchmark figures as validated results. They are not reproducible from the released artifact, and the corrections described here change what the numbers measure. What is offered instead is a system in which the benchmark can be run under a protocol that is stated precisely, checked automatically, and traceable to a recorded commit. For a field in which a plausible-looking image can be wrong in ways that matter clinically, that seems the more useful contribution.

Future work

Extending to multi-coil data with jointly estimated sensitivity maps is the natural next step, as is training across a wider range of acceleration factors than the two used here. Calibration of the uncertainty maps against actual reconstruction error would make them clinically interpretable rather than merely suggestive. Finally, the benchmark runs themselves remain to be executed on the corrected pipeline; the protocol in Section 8.3 specifies exactly what that requires.

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